Faculty of Information Technology

FIT3161 Computer Science Project Part 1
FIT3163 Data Science Project Part 1

Assessment details.
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Version 15042025

Project Initial Concept and Design/Data Analysis Report & Oral Presentation (Group submission)

This assessment task consists of 2 parts

A. Written submission worth 10% due Friday week 9, 11:55 pm

B. Oral presentation worth 15% due in weeks 10 and 11 during scheduled seminar or studio session

Unit Learning Outcomes.

1. Construct a project proposal.
2. Evaluate research methods and techniques of data collection and analysis to select those appropriate for a particular project.
3. Develop effective collaboration skills to operate effectively as a member of a development and research team.
4. Communicate the deliverables of a project in written and oral form to an appropriate audience.

Purpose of this assignment

1. Conceive your project goals and identify the components/parts required to develop the final software.
2. Communicate the project goals, establish and a broad set of requirements the initial design of your project. (Requirements will be refined later during project execution.)
3. Develop an initial understanding or overall vision of your final software that is shared by all team members. The ideas developed here will evolve as your team modifies and updates the project plan and design.
4. Provide the starting point for your team to build upon as you progress through the development phase.

1. Written submission report (10%)

Assignment specification

Start by clearly explaining the main goals of your project. Include a brief problem or opportunity statement, perhaps based on a case study, describing what you are expecting to deliver at the completion of the project. Use the course videos and notes as a reference to guide you in understanding the difference between opportunity statements and problem statements.

The team will provide a preliminary design for the software product that you will develop during the project (most development work occurs in your second project semester). In developing your design, your team must:

1. Using the course video materials as a guide to get you started, identify the software components or blocks (e.g., UI, back-end etc.). You do not need a lot of detail, two to three levels of decomposition is sufficient.

a. Identify the main user interface, accounting for your software's users and/or the location where the software will be used.

b. Identify the main data that will be processed (a database schema may be useful here).

c. Identify the main data processing components and data flow across the system.

2. Establish how the software components will be related to one another.
3. Briefly consider relevant software non-functional requirements as described in the course videos and notes, such as quality, accuracy and reliability.

4. Identify and describe the development platform and user platform for your project and the deliverable software.

5. FIT3163 Data Science students **ONLY**, must also undertake a preliminary analysis of the data (or part of a dataset) that you are likely to use.

Depending on your specific project, each of the above points 1-5 may be of different importance. For example, some projects may involve more user interaction than others and hence will place more importance on the user interface. Work with your team and client to decide how much importance to allocate to different software components as part of setting your project scope.

At this early stage, many teams will not yet have a detailed understanding of their software requirements. Hence, several different designs and approaches may appear to be plausible. You may mention several possible alternatives in your submission and presentation as long as you indicate which criteria you are likely to use later to come to a decision about the approach you take.

Deliverables to include in your submission

1. **Design representation**

Computer Science Students (FIT3161)	**Data Science Students (FIT3163)**
Two representations of your proposed software project design. 1) One will be a UI or a web interface mockup diagram, **&,** 2) The other representation can be chosen from the list below.	One representation of your proposed software project design. This will be a UI or a web interface mockup diagram. **&,** An analysis of the data set (or a part of it) that you will use in your project. Further info. is in the ***data analysis report*** on Moodle, "Additional files".

List of (2nd) possible design representation (for FIT3161 students only)

A. Database schema
B. Flow chart
C. Structure chart or block diagram showing software modules
D. Pseudo-code, structured English

Ensure that the software representation(s) and data analysis are consistent. Therefore, all team members must work closely together. Components described may be static structural features, or dynamic processes carried out by the software.

2. **Software and hardware specification.**
A specification of the technical and software environment required for the system is required. This may include *options* for specialized software, software libraries, database system, operating software, programming language environment (IDE, compiler, etc.), hardware requirements, data storage requirements, networking requirements, computing requirements. Project management tools should also be included (e.g. revision control system, code bug tracker etc.).

Ensure that you justify all choices.

Note here, you only need to justify the choices for the software and hardware requirements. You DO NOT need to justify or explain your representation(s). You just need to show them. The justifications and explanations of your representation(s) will be covered in your oral presentation.

It is recognized that at this stage of your project you may be simply guessing about many aspects of how it will unfold. You may have only a vague understanding of your project's technical requirements. You should still provide reasonable justification for your decisions and choices, and identify options that are plausible, as well as ruling out implausible or unsuitable options. It is acceptable to present alternatives and suggest how a final decision could be made. Awareness and discussion of potential software and system performance and quality issues will be viewed favorably.

Submission formats

Written submission: due Friday week 9, 11:55 pm
Submit a text document in pdf format. Include a title page, table of contents, and bibliography of references used. All written submissions must be structured and formatted to a professional level.

Team Members' Contribution Declaration
Team members' contributions must be declared in the Team Members Contribution declaration spreadsheet.

IMPORTANT: Failure to include a final Team Members' Contribution Declaration will incur an automatic 10% penalty on the final mark for the assessment.
Make sure that ALL team members need to click the SUBMIT button for the submission to be marked for grading.

1. One week before submission or earlier: pre-declaration
    a. Download the contribution declaration spreadsheet from Moodle.
    b. Fill in the pre-declaration with your planned work allocations, and submit it on Moodle at least 1 week before the assessment submission date.
2. On submission day: Final declaration
    a. Update the pre-declaration spreadsheet with the actual contributions you made.
    b. Submit the updated declaration form on Moodle on the 'FINAL- Team contribution (Project Concept and Design/Data Analysis report-written)' link by the main submission deadline.

2. Oral presentation(15%) due in weeks 10 or 11 in seminar/studio sessions

Assignment Specification

The oral presentation is a team task in which all team members must participate equally. It must be coherent, well-organised and professionally presented.

Orally present your initial project concept and design in a coherent, well-structured, professional way. All members are expected to participate equally in the presentation. Each member is expected to answer at least one question from the marking panel or contribute to a team answer.

Presentations will be scheduled during studio and seminar sessions of weeks 10 and 11. A presentation schedule will be made available.

There is no specified presentation structure, although the following format may be used **as a guide**.

- An introduction. This can include the main goals of your project or a brief opportunity statement.
- Design representations:

Computer Science Students (FIT3161)	**Data Science Students (FIT3163)**
1) Include both the representations from your written report. For each representation, justify and explain the overview, the software components and the relationships they show, **&**, 2) Requirements Traceability Matrix (RTM) *(See details below)*	1) Your representation of the UI or web interface mockup diagram from your written report. Justify and explain the overview, the software components and the relationships they show, **&,** 2) A summary of your data analysis component, including any visual and data-related materials, **&,** 3) Requirements Traceability Matrix(RTM) *(See details below)*

Include and discuss a comprehensive **Requirement Traceability Matrix (RTM).**
The RTM should clearly outline both functional and non-functional requirements and must include the following columns: , Requirement Description, Requirement Category, Source, and Current Status. The matrix should demonstrate how each requirement is traced throughout the project lifecycle, ensuring alignment with project goals and stakeholder expectations.

- “Other business” (an opportunity for you to raise any other relevant matters you wish - optional).
- A conclusion
- A bibliography (if required)

Organise your presentation content to communicate most effectively.

Presentation format

All presentations will be made on campus. All team members must contribute equally to the presentation and be present to answer questions.

Submission

Submit the following 2 documents (separately) on the same submission link:

1) Slides in **PDF format**, with file name << Team_No>>.pdf, need to be submitted on Moodle at the end of the presentation (For example: Team_CS_05.pdf).
2) Your requirements traceability matrix (RTM) in a separate document. The document should contain your team member's names, team number and your full RTM signed off by your supervisor.

Deadline for submission: **11.55 pm on the day of the presentation. Submit both the pdf slides and your RTM document on the same submission link.**
You can find the submission link on Moodle under the Assessments page.

Presentation requirements

1) Time for presentation

10 mins (team of 3) or
12 mins (team of 4) or
14 mins (team of 5) or
+ question time (length on a case-by-case basis at the discretion of tutors)

2) When showing slides, please clearly indicate the slide numbers and the name of the speaker(as shown on Allocate) at the top right-hand side of the slide that the speaker is talking about.

Other presentation suggestions

1) Balance your slide content with text and graphics as appropriate. Slide content should add value to the presentation and not distract the audience.
2) As far as possible, use at least 17pt font-size for the text on your slides.
3) Please ensure a seamless transition between presenters. You should rehearse the presentation a few times, especially the handover between presenters. Plan your presentation well among all team members. Exceeding the time limit will be penalised.
   a) Every 30 seconds over the time limit will incur a 5% penalty.
   b) The teaching staff can stop the presentation if the time is exceeded significantly.
4) Use cue cards or notes in point form/keywords to help remember your part of the presentation.
5) Avoid reading directly from your notes. Only read a sentence verbatim if it is a direct quote.
6) Use the Presenter Notes feature in Powerpoint/Google slides etc.
7) Step forward, face the audience, make eye contact, speak clearly and confidently.
8) Its better to have less content than too much content (You can always be asked for more details during question time)

An additional document containing oral presentation tips and suggestions is also available on Moodle, “Additional Information and Resources” section.

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Marking Criteria

Marking criteria for the written report part

Criteria	*Marks*
Title sheet (1) + Short introduction (4)	5
Project goals and expected deliverables	*10*
Representation (each)	*20*
Data analysis report (FIT3163 students only)	*20*
Software specification	*15*
Hardware specification	*15*
Justification of choices	*10*
Presentation + References	*5*
Total	***100***

Please note, there is also a detailed marking rubric under the additional files section on Moodle below the assignment specification.

Marking criteria for presentation

Each criterion will be marked out of 10. The overall presentation will be marked out of 60.

Criteria	**Marks**
Individual content	10
Individual presentation skills	10
Team presentation skill and coordination	10
Visual aids / slide presentation	10
Content relevance and completeness	10
Question and Answer responses and professionalism	10
Total	**60**

Please note, there is also a detailed marking rubric under the additional files section on Moodle below the assignment specification.

Important:
If your team cannot be present at the scheduled time for any reason, you must notify the teaching team as soon as possible to determine what assessment arrangements, if any, need to be made.

Further notes regarding the attendance to the presentation and special consideration if needed:

Failure to attend a presentation without either,

1) Pre- or post-presentation approved special consideration,
2) A course-related exemption such as a clash with another unit's assessment or a requirement to attend a formal, scheduled study-related activity outside of this unit, will incur the following "no show" penalties in addition to any late or other penalties applied to the assessment:

* without a warning to teaching staff at least 24 hours prior to the presentation scheduled time, penalty = 40% of available marks.
* with a warning to teaching staff at least 24 hours prior to presentation, penalty = 20% of available marks.

Late Submission

1. Submission must be made by the due date. Late submission penalties will be deducted according to Monash University policy. (See Moodle, "Additional information and resources" section).
2. If you believe that your assignment will be delayed because of circumstances beyond your control such as illness, apply for an extension prior to the due date. Extensions and individual alterations to the assessment regime will only be considered using the University Special Consideration Policy (See Special Consideration website, and follow the procedure).

Use of Generative AI tools in Projects and Assignments
See this assessment's Moodle module for the policy regarding use of GenAI in this unit.